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### Turf aerator with covered tine assembly

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#### Abstract

A turf aerator has a chassis and a motor mounted on the chassis between a front end and a rear end of the chassis and between a first side and a second side of the chassis. A first shaft is rotatably driven by the motor, a first inner tine assembly mounted to a first portion of the first shaft, a second inner tine assembly mounted on the first shaft, and a first outer tine assembly mounted to a second portion of the first shaft. A tine cover is removably secured to the chassis over at least one of the first inner tine assembly and the first outer tine assembly.

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## Background/Summary

CROSS REFERENCE (1) The present application is a divisional application of U.S. application Ser. No. 17/207,840, filed Mar. 22, 2021, which is a divisional application of U.S. application Ser. No. 16/284,167, filed on Feb. 25, 2019, which is a divisional application of U.S. application Ser. No. 15/331,931, filed on Oct. 24, 2016, which is a divisional application of U.S. application Ser. No. 13/282,095, filed on Oct. 26, 2011, which claims benefit to U.S. Provisional Application No. 61/407,126, filed on Oct. 27, 2010.

### BACKGROUND

(1) Methods and apparatus for aerating turf are shown and described.

(2) The conception of a pair of tine shafts driven through a differential shaft in a turf aerator as taught in U.S. Pat. No. 6,708,773, which is hereby incorporated herein by reference, represented a major innovation in the turf field. In particular, the ability to operatively connect and clutch one or the other or both of the tine shafts, whether done mechanically, hydrostatically, or in other manners or combinations, allow the operator to make turns while the tines are still in contact with the turf. However, once turning was a possibility, operators desired further ability in directing the turf aerator during operation.

(3) As an example, in an alternate approach by others to this major innovation, outer tine assemblies were rotatably mounted at the two end portions of a single driven shaft including inner tine assemblies secured to the single driven shaft intermediate the outer tine assemblies. Although increasing turning ability relative to conventional turf aerators existing prior to U.S. Pat. No. 6,708,773, this alternate approach did not have the turning ability achieved by the turf aerator of U.S. Pat. No. 6,708,773.

(4) Further, during normal operation of the turf aerator, the tines need servicing such as the result of blockage and breakage. Such servicing was performed from underneath the conventional turf aerator which was often held in a tilted condition. Furthermore, replacement of tines was difficult to accomplish by the operator, and tine reel assemblies utilized were difficult and expensive to manufacture and assemble.

(5) Likewise, to provide penetration of the tines into the turf, weights are removably attached to the sides of the chassis in conventional turf aerators. It can then be appreciated that such removable weights increased the overall mass which resulted in increased operator fatigue and apparatus wear as well as reduced maneuverability.

(6) Additionally, conventional turf aerators utilized various arrangements to position transport wheels relative to the chassis to allow transport without tine engagement as well as to control the depth of tine penetration. However, such arrangements were relatively complicated in design increasing cost and maintenance as well as requiring tools for adjustment.

(7) Thus, need for improvement in methods and apparatus for aerating exists in the field of turf management.

### SUMMARY

(8) This need and other problems in the field of turf management are solved by providing an outer tine assembly rotatably mounted adjacent an outer end of each of first and second shafts each having an inner tine assembly secured thereto for rotation therewith. A turf aerator independently driving the first and second shafts, such as by a hydrostatic transaxle, has enhanced and unexpected ability to be directed by operators during operation and while the tine assemblies are in contact

with the turf.

(9) Further, each of a plurality of tines of the tine assembly are secured by a single bolt extending diametrically through each tine located in a cavity of a tine plate, with the plurality of cavities being arranged in a non-intersecting manner in the tine plate. The tine plates of the inner tine assemblies are connected directly or through an attachment ring to the shafts, while the tine plates of the outer tine assemblies are connected directly or through an attachment ring to a sleeve rotatable on the shafts.

(10) Additionally, a tine cover is removably secured to the chassis and closes an opening in the top plate of the chassis positioned over the tine assembly of the turf aerator. Maintenance and replacement of the tine assembly can be accomplished, after removing the tine cover, through the opening. Thus, tilting the chassis to provide access from underneath the tine assembly as required in conventional turf equipment is avoided.

(11) Furthermore, weights are slideably received in a tine cover located over the tine assembly to be positioned intermediate the first and second side plates of the chassis and over the tine assembly. Positioning the removable weights generally centered over the tine assembly provides increased leverage which was not obtained by conventional turf aerators. Also, the weights can be removed from the tine cover if the tine cover is removed from the chassis such as for maintaining or servicing the tine assembly.

(12) In other aspects, transport wheels are mounted on ends of arms having their opposite ends pivotally mounted to the chassis. Abutment bars are pivotally mounted to the arms and to stop plates pivotally mounted to the chassis. The transport wheels can be pivoted between transport and operational positions where the upper pivot axes of the abutment bars are located in front or behind of the lower pivot axes of the abutment bars and the pivot axes of the stop plates. Pivoting of the stop plates can be easily accomplished such as by lift tubes pivotally connected to the stop plates and a lift handle pivotally mounted to the handlebar. The extent that the transport wheels move towards the chassis is limited by stops connected to the stop plates and abutting with the abutment bars in the operational position. Thus, the extent can be adjusted easily without the use of tools by simply changing the position of the stops in the stop plates.

(13) Illustrative embodiments will become clearer in light of the following detailed description in connection with the drawings.

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## Description

### DESCRIPTION OF THE DRAWINGS

(1) The illustrative embodiments may best be described by reference to the accompanying drawings where:

(2) FIG. 1 shows a front, perspective view of an embodiment of an apparatus for aerating turf.

(3) FIG. 2 shows a rear, perspective view of another embodiment of an apparatus for aerating turf.

(4) FIG. 3 shows an exploded perspective view of the chassis, transaxle, V-belt and front wheels of the aerating apparatus of FIGS. 1 and 2.

(5) FIG. 4 shows an exploded perspective view of the tine reel assemblies of the aerating apparatus of FIG. 1.

(6) FIG. 5 shows an exploded perspective view of the tine reel assemblies of the aerating apparatus of FIG. 2.

(7) FIG. 6 shows a partial cross sectional view of the aerating apparatus of FIG. 2.

(8) FIGS. 7A, 7B and 7C show partial side views of the aerating apparatus of FIG. 2.

(9) All figures are drawn for ease of explanation of the basic teachings only; the extensions of the figures with respect to number, position, relationship, and dimensions of the parts to form the illustrative embodiments will be explained or will be within the skill of the art after the following

description has been read and understood. Further, the exact dimensions and dimensional proportions to conform to specific force, weight, strength, and similar requirements will likewise be within the skill of the art after the following description has been read and understood.

(10) Where used in the various figures of the drawings, the same numerals designate the same or similar parts. Furthermore, when the terms “top”, “bottom”, “first”, “second”, “forward”, “rearward”, “reverse”, “front”, “back”, “height”, “width”, “length”, “end”, “side”, “horizontal”, “vertical”, and similar terms are used herein, it should be understood that these terms have reference only to the structure shown in the drawings as it would appear to a person viewing the drawings and are utilized only to facilitate describing the illustrative embodiments.

#### DESCRIPTION OF THE ILLUSTRATIVE EMBODIMENTS

(11) Illustrative embodiments of an apparatus for aerating turf are shown in the drawings and generally designated **10** and **10'**. Generally, turf aerator **10** and **10'** includes a chassis **12** including first and second side plates **14**, front and back plates **24** and **26** extending between side plates **14**, and a top plate **28** extending between side plates **14** and between front and back plates **24** and **26**. In the form shown, top plate **28** is generally rectangular. Chassis **12** is moveably supported on turf adjacent to front plate **24** by wheels **30** at opposite ends of a hydrostatic transaxle **32**, with wheels **30** located outwardly of side plates **14**. A handlebar **36** extends from chassis **12** in a direction away from front plate **24** and wheels **30**.

(12) A motor **38** is mounted to top plate **28** adjacent to front plate **24**. Motor **38** is in driving connection to hydrostatic transaxle **32** such as by a belt drive arrangement.

(13) In the embodiments shown, a pair of inner tine assemblies **100a** and **100b** are secured to a shaft **102** each driven at the outer end by a sprocket **18** and a roller chain by hydrostatic transaxle **32**. Floating, outer tine assembly **104** is rotatably supported on shaft **102**. It should be appreciated that tine assembly **104** is located axially outward of tine assemblies **100a** and **100b**. Specifically, inner tine assemblies **100a** and **100b** are secured to shaft **102** adjacent to its inner end and are rotatable with shaft **102**. Outer tine assembly **104** is rotatably mounted to shaft **102** and is located intermediate the inner tine assemblies **100a** and **100b** and the outer end of shaft **102**. Inner tine assembly **100b** is intermediate inner tine assembly **100a** and outer tine assembly **104**. Shaft **102** is driven at the outer end outwardly of outer tine assembly **104**. In the embodiments shown, first and second shafts **102** are independently rotatable about a tine axis with their inner ends located intermediate their outer ends and are independently driven at their outer ends outwardly of outer tine assemblies **104**. Each first and second shaft **102** includes tine assemblies **100a**, **100b** and **104** as described hereintofore. First and second shafts **102** and tine assemblies **100a**, **100b** and **104** are rotatably mounted to chassis **12** between side plates **14** and adjacent to back plate **26**. Shafts **102** and tine assemblies **100a** and **100b** are driven by motor **38** through hydrostatic transaxle **32**. This arrangement provides unexpected, surprising, and synergistic results compared to turf aerators which had all tine wheels driven or where the floating tine wheels were located intermediate the driven tine assemblies. In particular, providing outer tine assembly **104** rotatable adjacent only the outer end or end portion rather than adjacent to both ends or end portions of a single shaft **102** results in abilities to direct turf aerator **10** and **10'** during operation which was not previously contemplated and which was unexpected and surprising in the field.

(14) Further, in the embodiments shown, tine assemblies **100a**, **100b** and **104** each includes a circular tine plate **110** including a plurality of non-intersecting, generally semi-circular cross section cavities **112** extending along chords spaced from but parallel to diameters of the circular tine plate **110**. Each tine plate **110** includes an axial bore **114**.

(15) Each tine assembly **100a**, **100b** and **104** includes a plurality of tines **120**. Each tine **120** is formed by bending a flat sheet **122** of metal into a tubular shape. The abutting edges **124** of the flat sheet **122** each includes a cutout **125** to define a rear opening in the tubular shape. Each tine **120** is held in a cavity by a single bolt **126** extending through a through hole **128** extending diametrically through the tubular shape of the tine **120** and generally at 90° to abutting edges **124** and tine plate

**110.**

(16) In one embodiment, axial bores **114** of tine plates **110** of tine assemblies **100a** and **100b** are welded to shaft **102**, while axial bore **114** of tine plate **110** of tine assembly **104** is welded to a sleeve **116** rotatably mounted to shaft **102**. Each bolt **126** extends through an aperture **118** in tine plate **110**, extends through one of the cavities **112**, extends through through-hole **128** of tine **120** and through washers, and is threaded into a nut **129**.

(17) In an alternate embodiment, each tine assembly **100a**, **100b** and **104** includes a circular attachment ring **130**. Each bolt **126** extends through an aperture **132** in attachment ring **130**, through aperture **118** in tine plate **110**, through one of the cavities **112**, extends through through-hole **128** of tine **120** and through washers and is threaded into nut **129**. Attachment rings **130** of tine assemblies **100a** and **100b** are welded to shaft **102**, while attachment ring **130** of tine assembly **104** is welded to sleeve **116**. Thus, in this embodiment, tine plates **110** of tine assemblies **100a**, **100b** and **104** are free of attachment to shaft **102** and/or sleeve **116**.

(18) Thus, tine assemblies **100a**, **100b** and **104** can be easily manufactured and assembled and installed in turf aerator **10** as well as allow ease of replacement of tines **120** in the field. In particular, tines **120** are securely held in cavities **112** by a single bolt **129** and are completely exposed outside of cavities **112** without other components enclosing or sandwiching tines **120**. Thus, tines **120** can be easily removed by simply removing a single bolt **126**. Furthermore, tines **120**, tine plates **110** as well as attachment rings **130** are easy to manufacture and assemble.

(19) In the embodiments shown, chassis **12** of turf aerator **10** includes a rectangular opening **40** in top plate **28** spaced from side plates **14** and front and back plates **24** and **26** thereof and generally located over tine assemblies **100a**, **100b** and **104**. A tine cover **50** is removably secured to chassis **12** and closes opening **40** in chassis **12**. Tine cover **50** includes first and second side sheets **52** each having a seven sided cross section located outwardly of side plates **14** of chassis **12**. Thus, tine cover **50** can be removably secured to chassis **12** by fasteners such as screws **54** extending through side sheets of tine cover **50** and threaded in or otherwise connected to side plates **14** of chassis **12**. Further, front and back sheets **56** and **58** of tine cover **50** abut with top plate **28** of chassis **12** outwardly of opening **40**. Tine cover **50** further includes a top sheet **60** extending between front and back sheets **56** and **58** and extending between side sheets **52**. Front sheet **56** includes a first section **56a** extending at an obtuse angle from top sheet **60** and generally perpendicular to side sheets **52**. Front sheet **56** further includes a second section **56b** extending from first section **56a** generally perpendicular to side sheets **52** and parallel to and spaced from back sheet **58**. Top sheet **60**, side sheets **52**, and back sheet **58** extend generally perpendicular to each other. Front and back sheets **56** and **58** abut with top plate **28** outwardly of opening **40**. Side sheets **52** each have seven edges including an edge **52a** interconnected to top sheet **60**, an edge **52b** interconnected to back sheet **58**, edges **52c** and **52d** interconnected to first and second sections **52a** and **52b**, an edge **52e** extending at an obtuse angle from edge **52d** to a bottom edge **52f** extending generally parallel to edge **52a**, and an edge **52g** extending at an obtuse angle from bottom edge **52f** to edge **52b**.

(20) Removable weights **70** are received in cavities **72** formed in tine cover **50**. In an embodiment shown, entrances **74** to cavities **72** are formed in side sheets **52** at the intersection of top sheet **60** and side sheet **52** and of back sheet **58** and side sheet **52**. In an alternate embodiment, an entrance **76** to cavity **72** is formed in back sheet **58**.

(21) It should be appreciated that placement of removable weights **70** in removable tine cover **50** allows their placement to be generally centered over tine assemblies **100a**, **100b** and **104** to provide increased leverage which was not obtained by conventional turf aerators. Thus, increasing leverage allows more efficient use of the mass of removable weights **70** ensuring penetration of tines **120** into turf, and resulting in decreased operator fatigue and apparatus wear as well as increase maneuverability. Furthermore, removal of tine cover **50**, preferably after removable weights **70** are removed therefrom, allows servicing of tine assemblies **100a**, **100b** and **104** from above and particularly not from below as in conventional turf aerators. Thus, tilting chassis **12** to provide

access from underneath tine assemblies **100a**, **100b** and **104** as well as the inherent risk of an operator working beneath the turf aerators **10** and **10'** as required in conventional turf equipment is avoided.

(22) Turf aerator **10** and **10'** can be configured in a transport position where transport wheels **134** are pivoted forward and underneath chassis **12**. Specifically, each transport wheel **134** is rotatably mounted to an end of an arm **136** about a wheel axis, with arms **136** pivotally mounted by their opposite ends to chassis **12** about an arm axis **138** intermediate front plate **24** and the tine axis defined by shafts **102** and parallel to and spaced from and parallel to the wheel axes of wheels **134**. Axis **138** is parallel to but spaced from the tine axis defined by shafts **102** and is located intermediate the tine axis defined by shafts **102** and front plate **24**. The spacings of the wheel axes of wheels **134** and the tine axis defined by shafts **102** from axis **138** are generally equal.

(23) A stop plate **140** is pivotally mounted to chassis **12** about a stop axis **142** spaced from and parallel to axis **138**, the tine axis defined by shafts **102** and the wheel axes of wheels **134** and located intermediate and vertically above axis **138** and the wheel axes of wheels **134**. Stop plate **140** is generally isosceles triangular shaped, with axis **142** located at the upper apex thereof. First and second threaded apertures **144** are located adjacent a lower apex of stop plate **140**. A stop **146** is removably connected to and moveable with stop plate **140** by being threadably received in one of the first and second threaded apertures **144** in the form shown.

(24) A lower end of an abutment bar **150** is pivotally mounted to arm **136** about a lower axis **150a** parallel to and intermediate axis **138** and the wheel axes of transport wheels **134**. Abutment arm **150** is further pivotally mounted to a lower apex of stop plate **140** about an upper axis **150b** parallel to and spaced from axes **138** and **142**, the wheel axes of transport wheels **134**, and lower axis **150a**.

(25) A lower end of a lift tube **160** is pivotally mounted to stop plate **140** about an axis parallel to and spaced from axis **142** intermediate and aligned with axis **142** and threaded apertures **144**. An upper end of lift tube **160** is pivotally mounted to a lift handle **162** in turn pivotally mounted to handlebar **36**.

(26) Abutment bar **150** and transport wheels **134** are movable between a transport position and an operational position. In the transport position, axis **150b** is located intermediate a plane extending through axes **142** and **150a** and transport wheels **134**. In the operational position, the plane extending through axes **142** and **150a** is located intermediate axis **150b** and transport wheels **134**. The extent that axis **150b** is spaced from the plane extending through axes **142** and **150a** is dependent upon abutment of stop **146** with abutment bar **150**. Thus, by simply moving stop **146** between either the first and second threaded apertures **144**, the extent that axis **150b** is spaced, and thus the spacing of transport wheels **134** from chassis **12**, is easily adjustable. Furthermore, since the force on stop **146** is in shear, stop **146** can be removed and replaced without the use of tools and in the form shown is simply rotated by the operator using fingers. Furthermore, due to the overcenter relationship of axis **150b**, there is no need for a spring bias of the lift arrangement. Thus, turf aerator **10** and **10'** includes a lifting arrangement which is very simple in design to thereby decrease cost and maintenance as well as allowing adjustment without the use of tools as described.

(27) Handlebar **36** is utilized to steer chassis **12** along the turf, or in other words to direct chassis **12** in turning to the right or left on the turf as well as to tilt chassis **12** relative to the turf about wheels **30** or **134**.

(28) Although shown utilized in a single machine and believed to obtain synergistic results, it should be appreciated that such features can be used alone or in other combinations. As an example, the tine assembly arrangement could be utilized with conventional tine assemblies and/or without removable tine cover **50** as in the form shown. Similarly, the removable tine cover **50** could be utilized in conventional turf aerators with or without individually driven tine assemblies **100a** and **100b** and/or with conventional tine arrangements. Likewise, the tine assemblies **100a**, **100b** and **104** could be utilized in conventional turf aerators with or without individually driven tine assemblies and/or without removable tine cover **50** as in the form shown.

(29) Thus since the invention disclosed herein may be embodied in other specific forms without departing from the spirit or general characteristics thereof, some of which forms have been indicated, the embodiments described herein are to be considered in all respects illustrative and not restrictive. The scope of the invention is to be indicated by the appended claims, rather than by the foregoing description, and all changes which come within the meaning and range of equivalency of the claims are intended to be embraced therein.

## Claims

1. A turf aerator comprising: a chassis having a first side plate at a first side, a second side plate at a second side, a front end plate at a front end, and a rear end, with the chassis being movably supported on turf; a handlebar extending from the chassis in a direction away from the front end plate; a motor mounted on the chassis between the front end and the rear end and between the first side and the second side; a first shaft rotatably driven by the motor; a first inner tine assembly and at least a second inner tine assembly mounted to a first portion of the first shaft; a first outer tine assembly mounted to a second portion of the first shaft; and a tine cover removably secured to the chassis over at least one of the first inner tine assembly and the first outer tine assembly.
2. The turf aerator of claim 1, wherein at least one of the first inner tine assembly, the second inner tine assembly, and the first outer tine assembly comprises: a tine plate including a plurality of non-intersecting, generally semi-circular cross section cavities; a plurality of tubular tines, with each tubular tine received in one of the plurality of cavities; and a bolt extending diametrically through each tubular tine, extending through the one of the plurality of cavities, and connected to the tine plate.
3. The turf aerator of claim 2, wherein each tubular tine is in a form of a flat sheet bent into a tubular shape and having abutting edges, with the abutting edges each including a cutout to define a rear opening in the tubular shape, with a through hole extending diametrically through the tubular shape of the tubular tine and generally at 90° to the abutting edges, with the bolt extending through the through hole, with the plurality of cavities extending along chords spaced from, but parallel to, diameters of the tine plate.
4. The turf aerator of claim 3 further comprising an attachment ring, with each bolt connected to the attachment ring, with the attachment ring welded to either the first shaft or a sleeve rotatably mounted on the first shaft, being free of direct attachment to either the first shaft or the sleeve.
5. The turf aerator of claim 1 further comprising: a removable weight, with the tine cover including a cavity slideably receiving the removable weight, with the removable weight received in the cavity extending over the tine assembly intermediate the first side and the second side.
6. The turf aerator of claim 5, wherein the tine cover includes a back sheet, a top sheet extending from the back sheet, and first and second side sheets extending from the back sheet and the top sheet, with the first and second side sheets located outwardly of and extending over the first and second sides, with the tine cover being removably secured to the chassis by fasteners extending between the first and second side sheets and the first and second side plates.
7. The turf aerator of claim 6, further comprising a top plate extending between the first and second side plates, and an opening formed in the top plate, wherein the opening is spaced from the first and second side plates and from the front and back walls; wherein the back, top, and side sheets extend perpendicular to each other; wherein the tine cover includes a front sheet including a first section extending at an obtuse angle from the top sheet and generally perpendicular to the first and second side sheets; wherein the front sheet includes a second section extending from the first section, generally perpendicular to the first and second side sheets, and parallel to and spaced from the back sheet, with the first and second side sheets having seven edges including edges interconnected to the back and top sheets and the first and second sections, a front edge extending at obtuse angle from the interconnection of the second section and the first and second side sheets to a bottom edge



spaced from and parallel to the top sheet, and a back edge extending at an obtuse angle from the interconnection of the back sheet and the first and second side sheets to the bottom edge.

8. The turf aerator of claim 7 wherein an entrance to the cavity is located in one of the first and second side sheets of the cover.

9. The turf aerator of claim 7 wherein an entrance to the cavity is located in the back sheet.

10. The turf aerator of claim 1, further comprising: a second shaft rotatably driven by the motor; at least a third inner tine assembly mounted to a first portion of the second shaft; and a second outer tine assembly mounted to a second portion of the first shaft.

11. The turf aerator of claim 10, wherein the second shaft is driven by the motor independently of the first shaft.

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